

(e) Find the values of x and y , given that x and y are real, and $\frac{1}{x+iy} - \frac{1}{1+i} = 1 - 2i$.

$$1 + i - x - yi = (1 - 2i)(1 + i)(x + iy)$$

$$1 + i - x - yi = (1 + i - 2i + 2)(x + yi)$$

$$1 + i - x - yi = (3 - i)(x + yi)$$

$$1 + i - x - yi = 3x + 3yi - xi + y$$

$$0 = 1 - y - 4x + (x - 4y + 1)i$$

$$1 - y - 4x = 0$$

$$y = 1 - 4x$$

$$x - 4y + 1 = 0$$

$$x - 4(1 - 4x) + 1 = 0$$

$$x - 4 + 16x + 1 = 0$$

$$17x = 3$$

$$x = \frac{3}{17}$$

$$y = \frac{5}{17}$$